Chapter 1 - Basics

This chapter introduces concurrency basics. Threads share memory. A thread may cache values, so one thread's write may not be visible to another.

Chapter 2 - Visibility

Visibility means a write by one thread becomes visible to others. Without proper synchronization, the JVM may reorder instructions. Visibility means a write by one thread becomes visible to others. Without proper synchronization, the JVM may reorder instructions. Visibility means a write by one thread becomes visible to others. Without proper synchronization, the JVM may reorder instructions. Visibility means a write by one thread becomes visible to others. Without proper synchronization, the JVM may reorder instructions. Visibility means a write by one thread becomes visible to others. Without proper synchronization, the JVM may reorder instructions. Visibility means a write by one thread becomes visible to others. Without proper synchronization, the JVM may reorder instructions. Visibility means a write by one thread becomes visible to others. Without proper synchronization, the JVM may reorder instructions.

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Chapter 5 - volatile and atomicity

volatile only guarantees visibility and ordering; it does NOT guarantee atomicity. The operation i++ consists of three steps: read i, add one, write back i. Even if i is declared volatile, two threads can interleave these three steps and lose updates. For example, two threads both read i=5, both write back 6, and the expected 7 is lost. To make i++ atomic, use synchronized or java.util.concurrent.atomic.AtomicInteger.

volatile vs synchronized:
synchronized guarantees atomicity, visibility and ordering, and is a mutual-exclusion lock.
volatile only guarantees visibility and ordering, is lightweight, and does not block threads.

Chapter 6 - Running plan

My running plan: run 5 km on Monday, Wednesday and Friday morning; long slow run 10 km on weekends. Do not run if it rains or air quality is bad.